Multiple regression models

2024 Semester 2

Multiple regression models, masked associations & Causal blocks

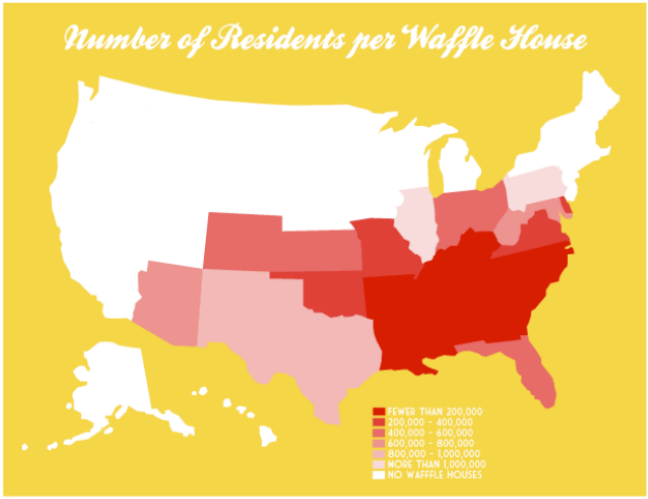
This week's goal:

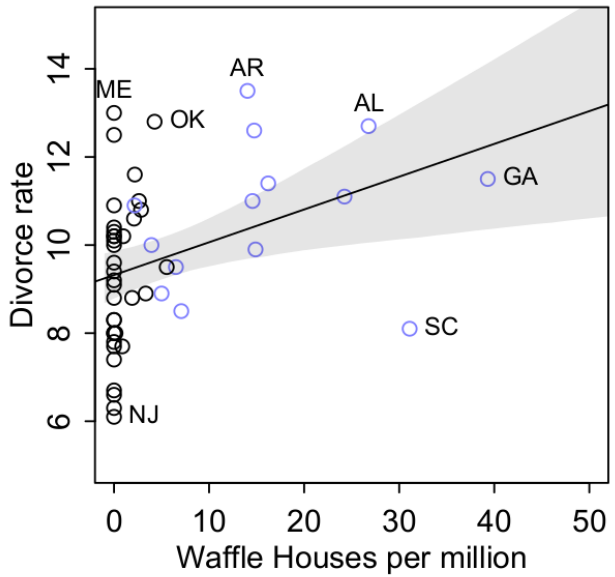
- Explore multiple regressions for masked associations
- Define and explore elementary confounding blocks

Reminders

- Causes are not in the data; interpreting parameter estimates depends on what we believe the causal model is (DAG).
- We will see that several (sometimes very many) causal models can be consistent with one set of parameter estimates.
- Good to get you to think about simulating causal effects in many different scenarios, incl. with interventions. See Section 5.1.5.3 of the SR book.

Multiple regressions





We saw in the divorce rate example that using multiple predictor variables helps knock out spurious associations.

We can also use multiple predictor variables to measure the direct influences of multiple factors.

Masked relationships

- Association between a predictor variable and an outcome can be masked.
- You need to observe both to see the "true" influence of either on the outcome.
- How do we account for the masking variable (seen as a nuisance)?

Masking situations tend to arise when:

- Both predictors are associated with one another.
- Have opposite relationships with the outcome



Milk and brain data example



Eulemur fulvus
0.49 *kcal/g*
55% *neocortex*

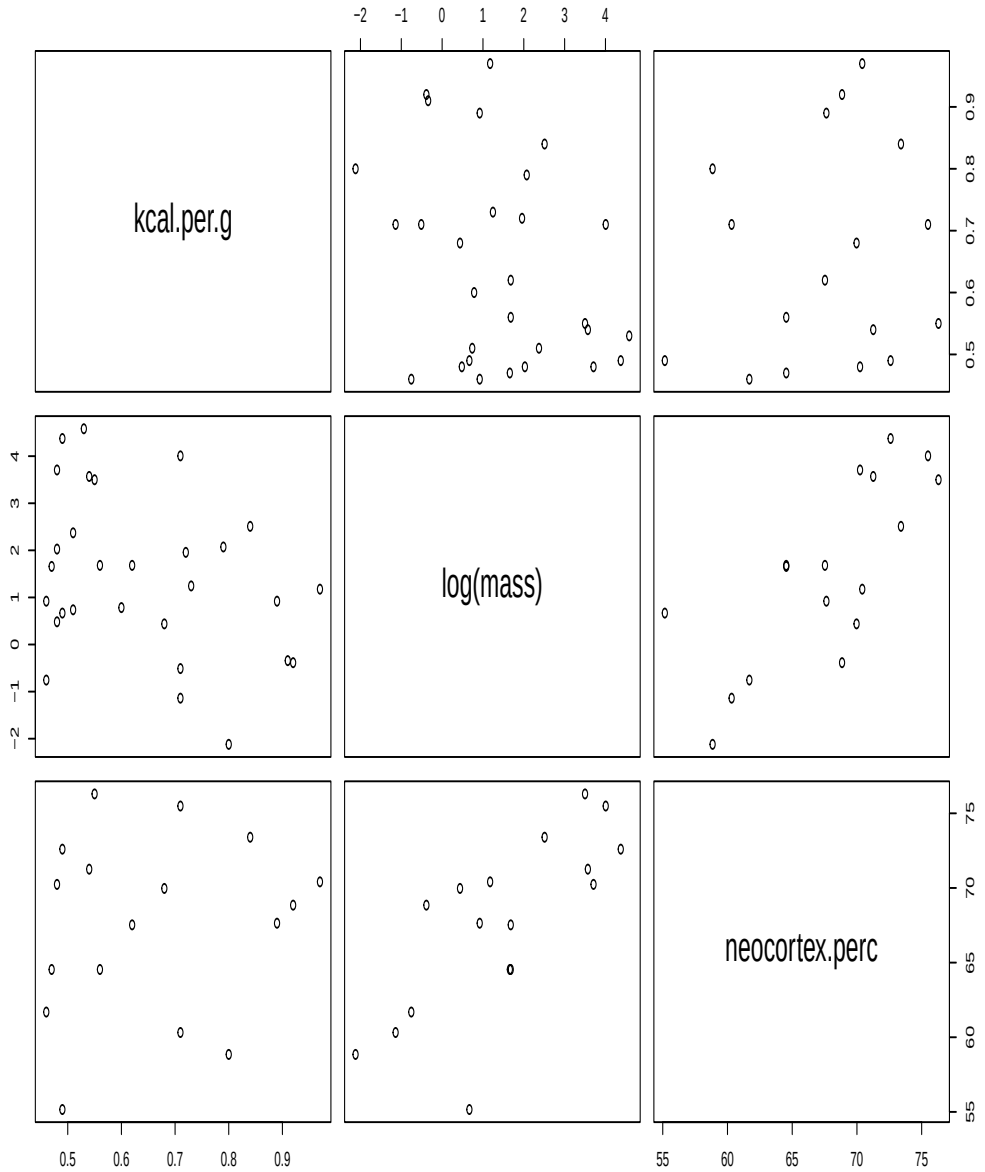


Homo sapiens
0.71 *kcal/g*
75% *neocortex*



Cebus apella
0.89 *kcal/g*
68% *neocortex*

Primate milk data




```
library(rethinking)
data(milk)
d<-milk
pairs( kcal.per.g +
      log(mass) +
      neocortex.perc,
      data = d)
```

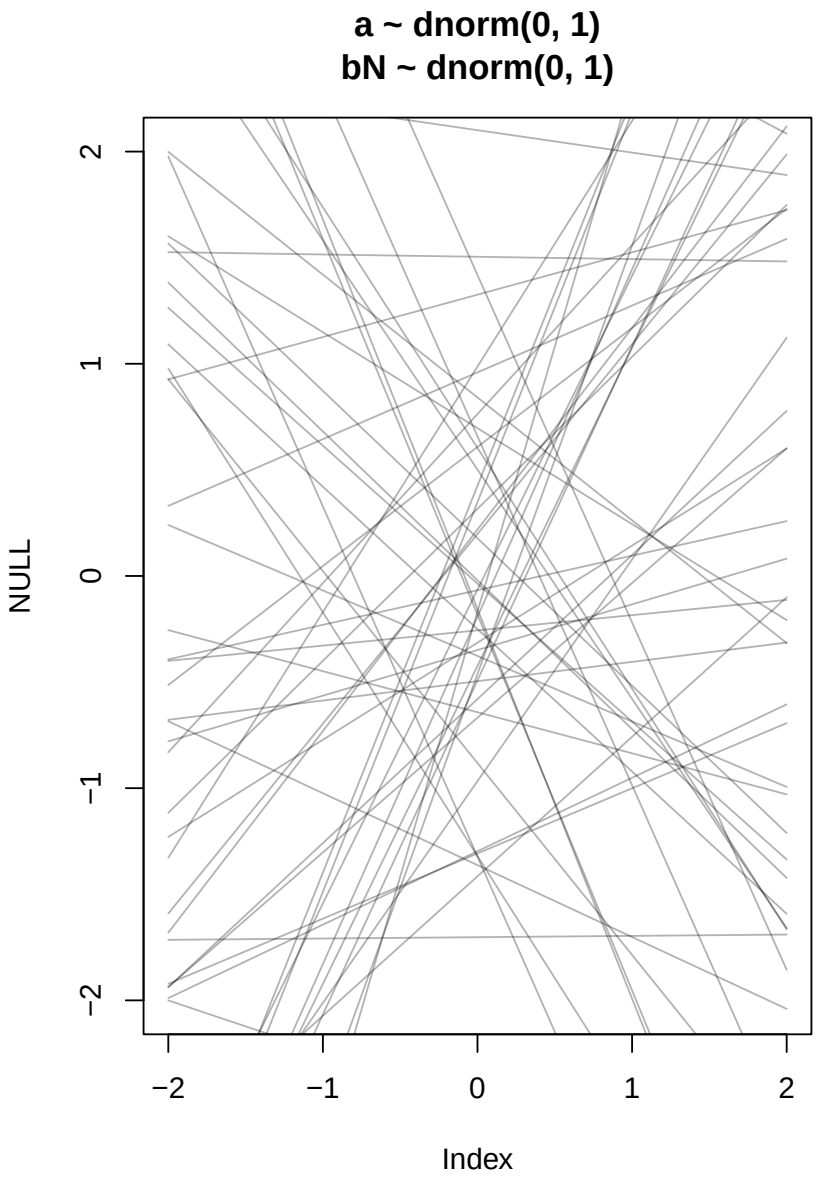
Bivariate model

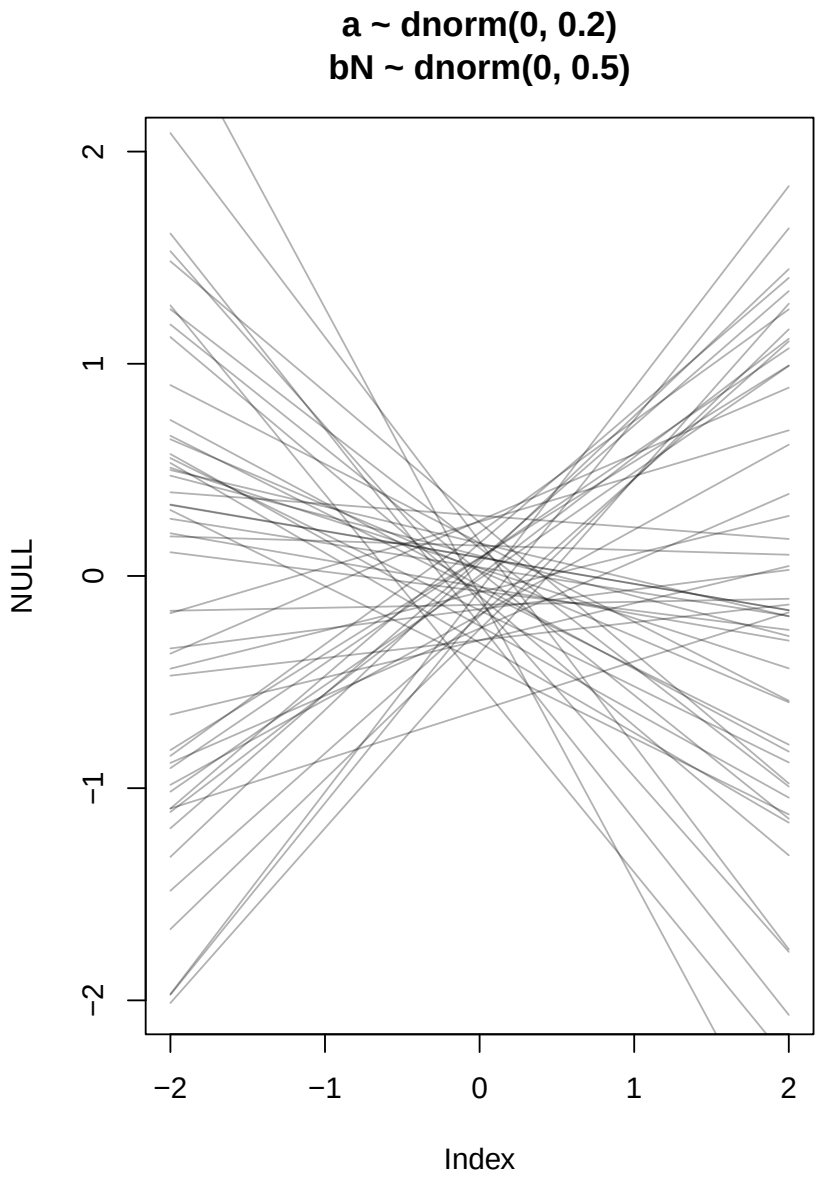
First we will consider simple bivariate regressions

$$
\begin{aligned}
K_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_N N_i \\
\alpha &\sim \text{Normal}(0, 0.2) \\
\beta_N &\sim \text{Normal}(0, 0.5) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$

$$
\begin{aligned}
K_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_M M_i \\
\alpha &\sim \text{Normal}(0, 0.2) \\
\beta_M &\sim \text{Normal}(0, 0.5) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$

Detour: reminder on priors

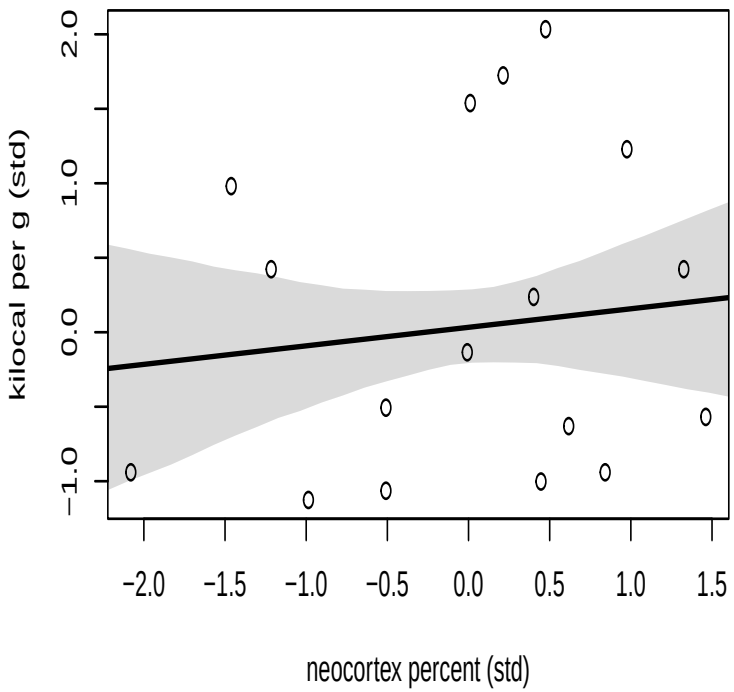


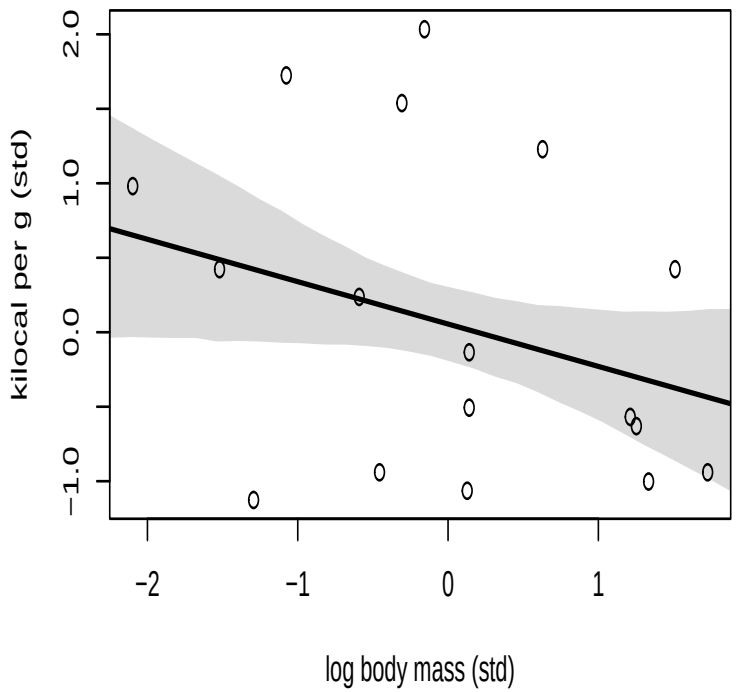


Bivariate regressions

$$
\begin{aligned}
K_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_N N_i \\
\alpha &\sim \text{Normal}(0, 0.2) \\
\beta_N &\sim \text{Normal}(0, 0.5) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$

$$
\begin{aligned}
K_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_M M_i \\
\alpha &\sim \text{Normal}(0, 0.2) \\
\beta_M &\sim \text{Normal}(0, 0.5) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$





Primate milk data modelling

```
m5.7 <- quap(alist(
    K ~ dnorm(mu, sigma),
    mu <- a+bN*N+bM*M,
    a ~ dnorm(0,0.2),
    bN ~ dnorm(0,0.5),
    bM ~ dnorm(0,0.5),
    sigma ~ dexp(1)),
data = dcc)
```

$$
\begin{aligned}
K_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_N N_i + \beta_M M_i \\
\alpha &\sim \text{Normal}(0, 0.2) \\
\beta_N &\sim \text{Normal}(0, 0.5) \\
\beta_M &\sim \text{Normal}(0, 0.5) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$

What happened?

```
precis(m5.5)
```

```
##             mean        sd       5.5%     94.5%
## a     0.03991976 0.1544906 -0.2069860 0.2868256
## bN    0.13323542 0.2237460 -0.2243539 0.4908247
## sigma 0.99981540 0.1647059  0.7365835 1.2630473
```

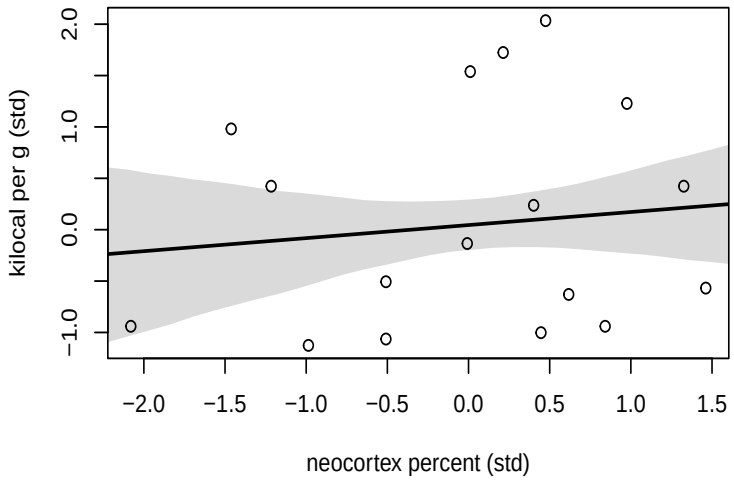
```
precis(m5.6)
```

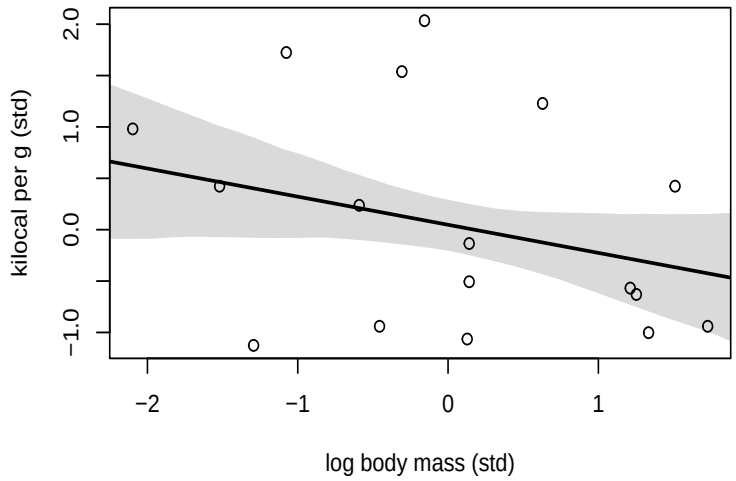
```
##              mean        sd       5.5%      94.5%
## a      0.04654149 0.1512800 -0.1952332 0.28831623
## bM    -0.28253576 0.1928818 -0.5907982 0.02572666
## sigma  0.94927964 0.1570616  0.6982648 1.20029444
```

```
# how does the effect of M and N change when they are both in the same model?
precis(m5.7)
```

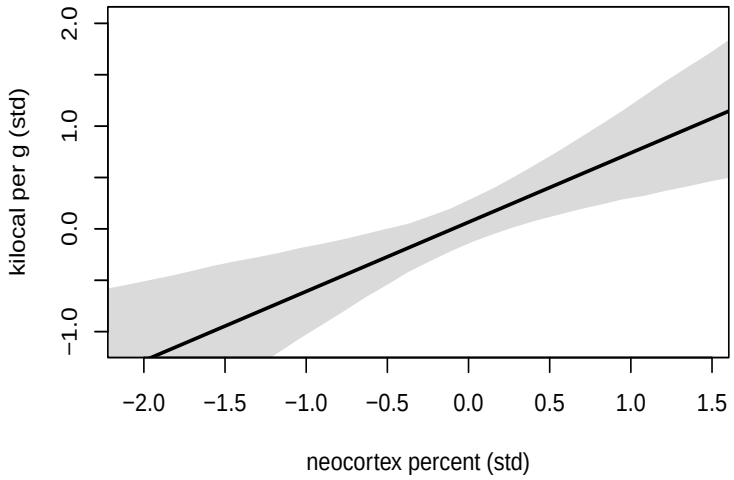
```
##              mean        sd       5.5%      94.5%
## a      0.06799257 0.1339986 -0.1461630  0.2821482
## bN     0.67511810 0.2482983  0.2782894  1.0719468
## bM    -0.70299149 0.2207867 -1.0558513 -0.3501316
## sigma  0.73801327 0.1324614  0.5263144  0.9497121
```

Bivariate vs Multiple regression analyses

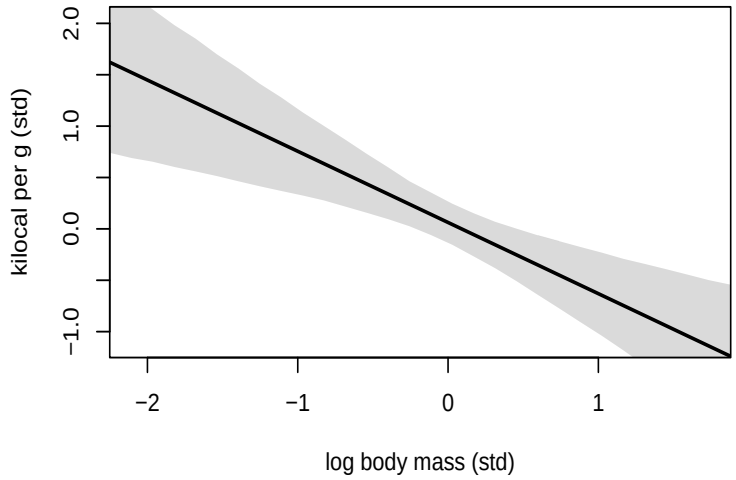




Counterfactual holding M = 0



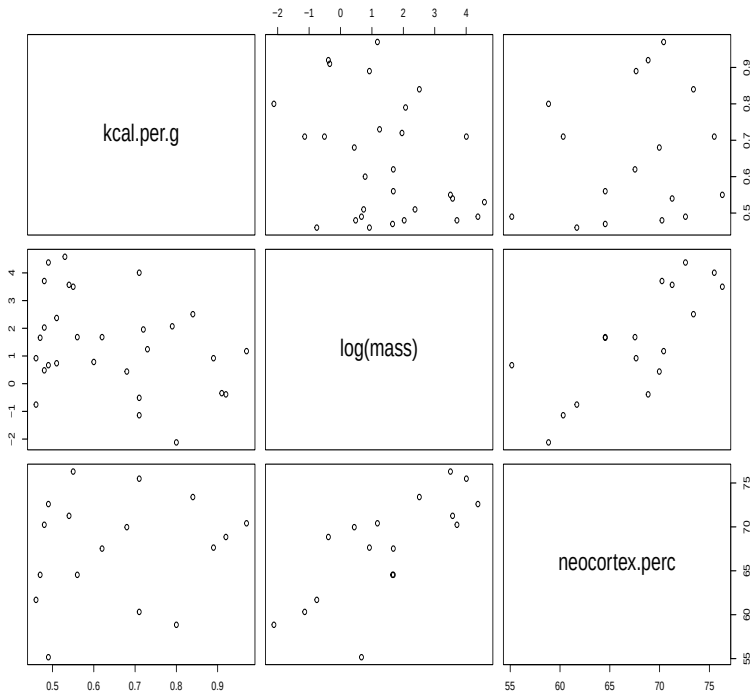
Counterfactual holding N = 0

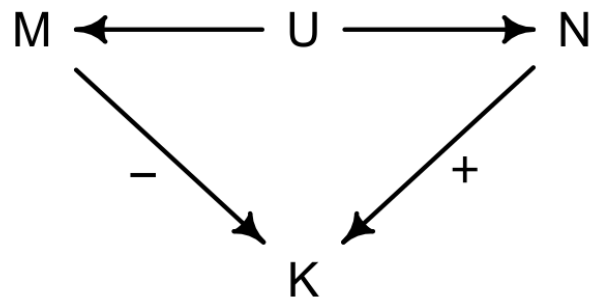


Masking situation

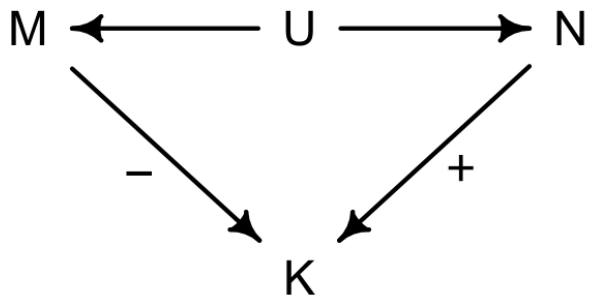
Tend to arise when

- Both predictors are associated with one another.
- Have opposite relationships with the outcome

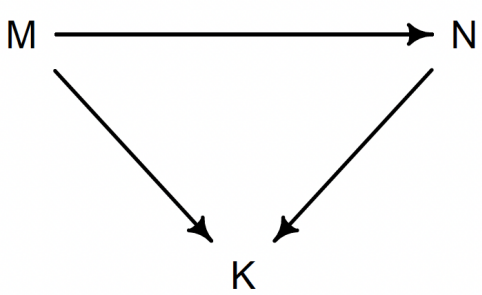


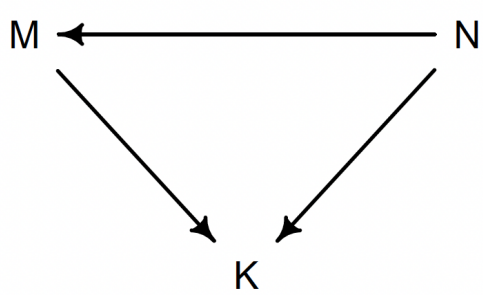


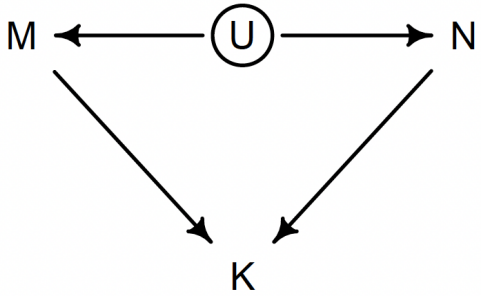
Multiple DAGs



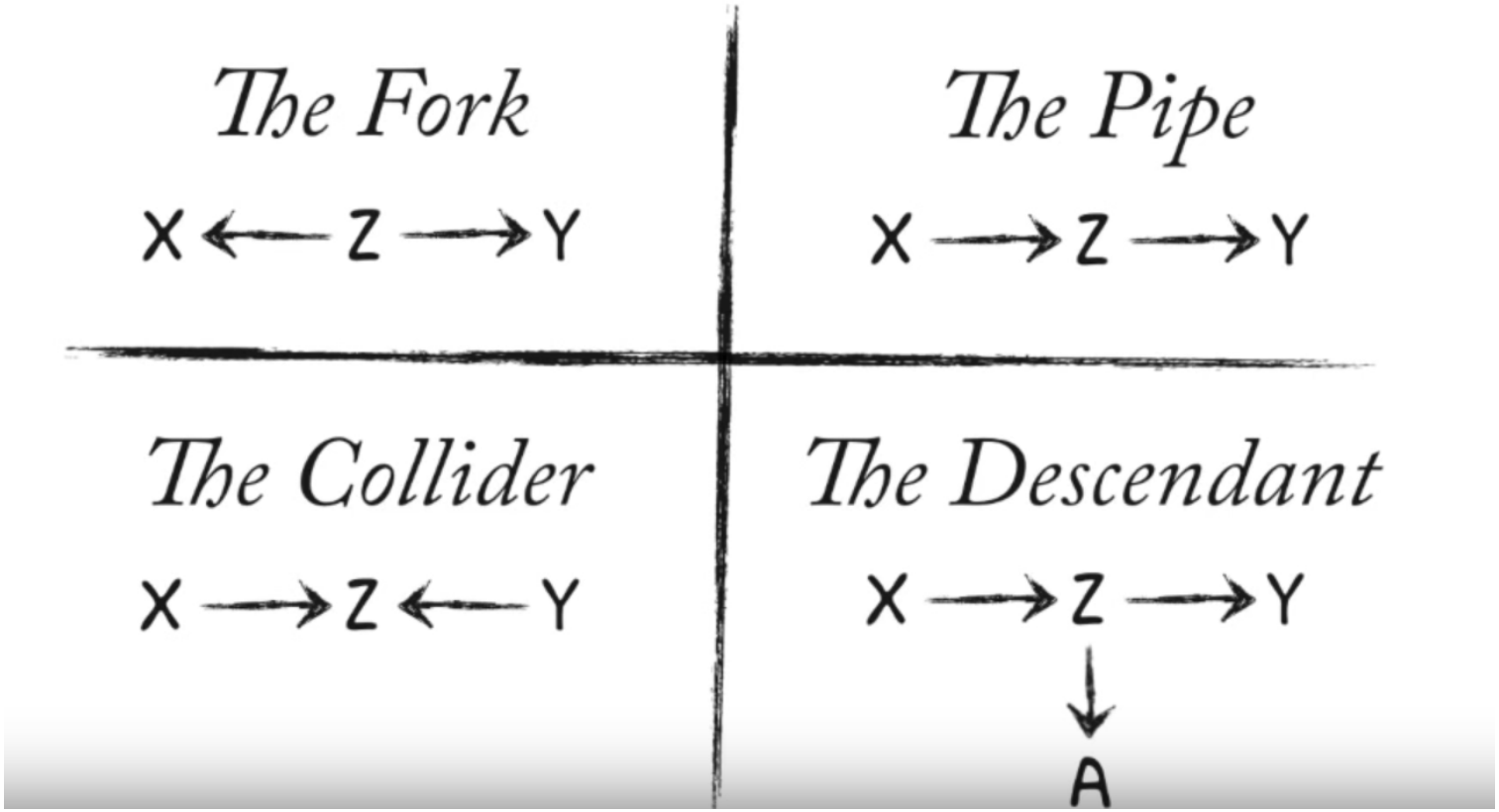
Cannot really distinguish DAG above to one with no U and either causal relationship between N and M; Markov equivalence







The causal cutlery: The four elementary confound blocks

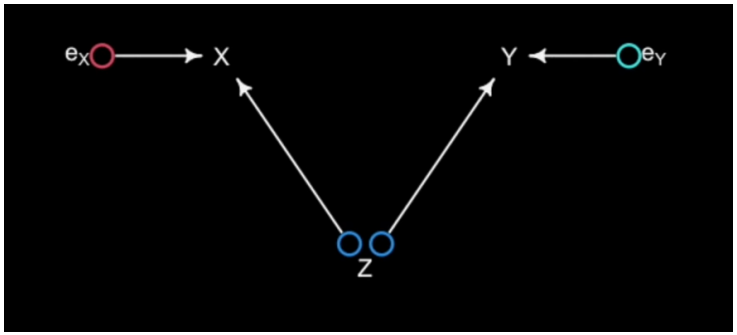


The "fork": Z is a confounder



- X and Y are associated: $X \not\perp Y$.
- if you have information about one, you have information about the other.
- They share a common cause.
- Once stratified by Z, we don't see the association between X and Y anymore: $X \perp Y | Z$.
- Stratifying by Z breaks the fork.

The fork: information flow and an example




```
n <- 1000
Z <- rbern(n, 0.5)
X <- rbern(n, (1-Z)*0.05+Z*0.95)
Y <- rbern(n, (1-Z)*0.2+Z*0.8)
```

```
table(X,Y)
```

```
##     Y
## X       0   1
##   0 392 135
##   1 115 358
```

```
cor(X,Y)
```

```
## [1] 0.5000225
```

The fork: example (cont.)

Now stratified by Z

```
table(X[Z==0], Y[Z==0])
```

```
##
##         0   1
##    0  388 111
##    1   14   3
```

```
cor(X[Z==0], Y[Z==0])
```

```
## [1] -0.01977992
```

Stratify based on a continuous variable

Need a function because now there are infinite values of Z

We now include it as a term in our model

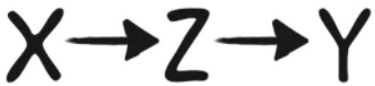
For example:

$$
\begin{aligned}
K_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_N N_i + \beta_M M_i \\
\alpha &\sim \text{Normal}(0, 0.2) \\
\beta_N &\sim \text{Normal}(0, 0.5) \\
\beta_M &\sim \text{Normal}(0, 0.5) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$

See "5.1.5 Plotting multivariate posteriors".

The pipe

Z is a mediator



- X and Y are associated: $X \not\perp Y$.
- The influence of X on Y is transmitted through Z.
- Once stratified by Z, we don't see the association between X and Y anymore: $X \perp Y | Z$.
- Stratifying by Z breaks the pipe.
- Note: statistically very similar to the fork but causally very different.

The pipe: a binary example

```
n <- 1000
X <- rbern(n, 0.5)
Z <- rbern(n, (1-X)*0.2+X*0.8)
Y <- rbern(n, (1-Z)*0.1+Z*0.9)

table(X,Y)
```

```
##     Y
## X      0   1
##    0 374 125
##    1 149 352
```

```
cor(X,Y)
```

```
## [1] 0.452572
```

```
table(X[Z==0], Y[Z==0])
```

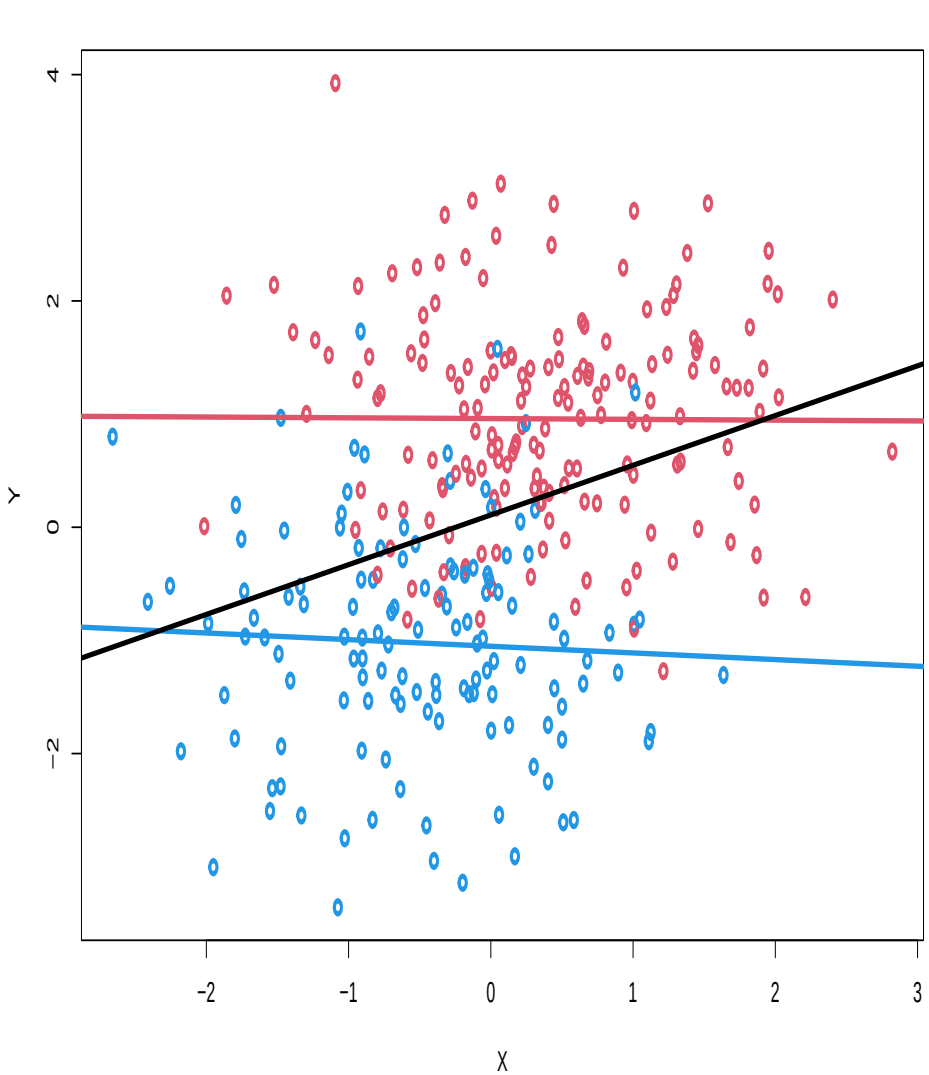
```
##
##         0   1
##    0  362  43
##    1   99  10
```

```
cor(X[Z==0], Y[Z==0])
```

```
## [1] -0.01939592
```

The pipe: a continuous example

```
n <- 300
X <- rnorm(n)
Z <- rbern(n, inv_logit(X))
Y <- rnorm(n, 2*Z-1)
```



Pipe real data example: Plant growth experiment

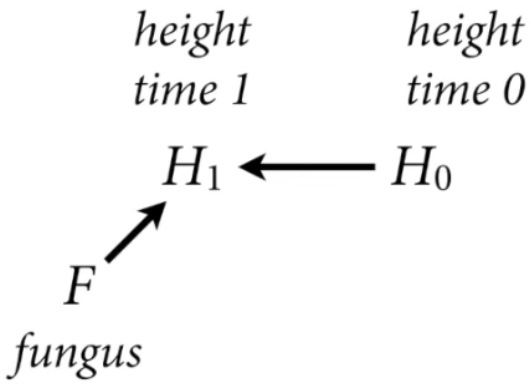
- An example of post-treatment bias.
- Shows that treatment of interventional experiments are not always synonymous to easy causal inference.
- 100 plants. Half treated with anti-fungal. We measure the growth (continuous, Δ between h_0 and h_1) and the fungus (binary).
- Estimand: causal effect of treatment on plant growth.

See Section 6.2 of the SR book.



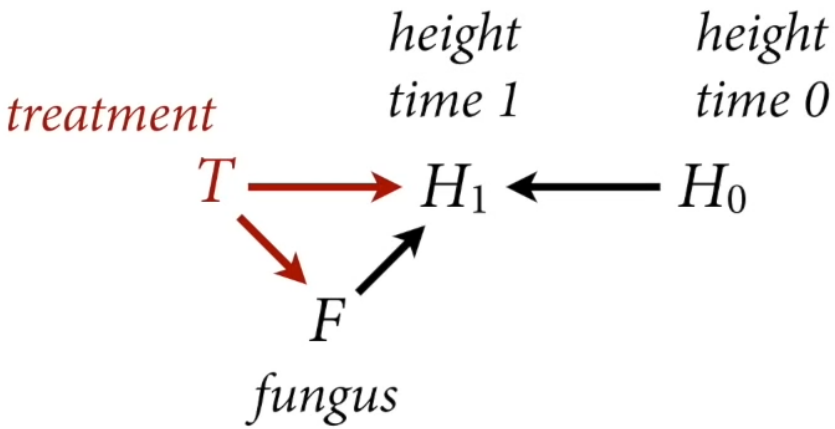
Plant growth treatment & fungus

The scientific model



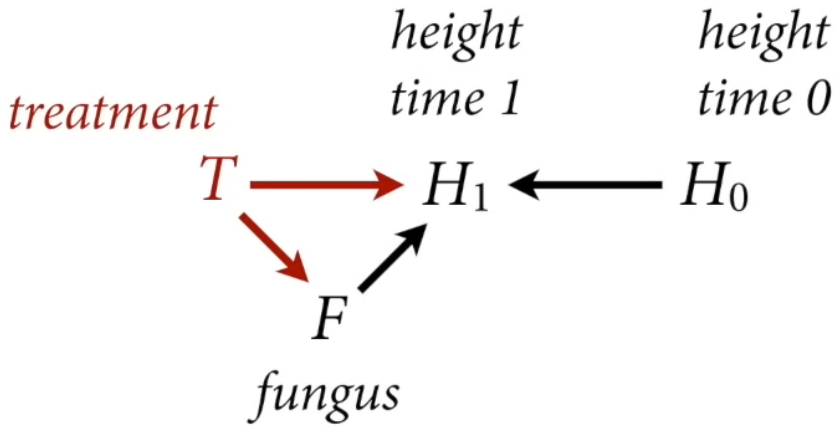
Plant growth treatment & fungus

The scientific model



Plant growth treatment & fungus

The scientific model



- Estimand: total causal effect of T on $\Delta(H_1 - H_0)$.
- $T \to F \to H_1$ is a pipe.
- Should we stratify by F?
- NO, it would block the pipe. See Section 6.2 of the SR book for details of the model construction, incl. simulations, a misleading model (full regression) and correct modelling approach.

Post-treatment bias

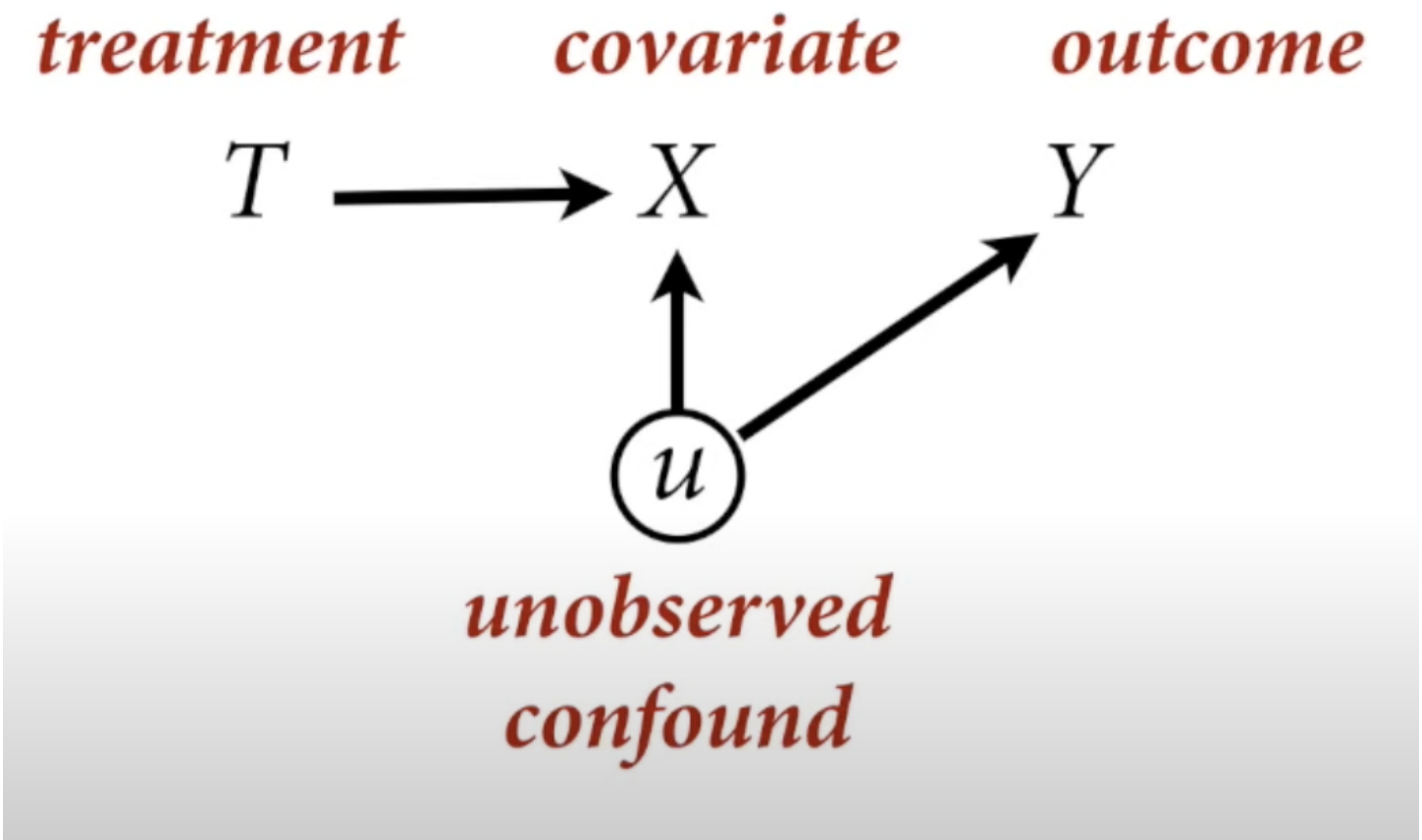
- Stratifying by (aka conditioning on) F induces *post-treatment bias*.
- Might conclude that treatment does not work when it actually does.
- Consequences of treatment should not usually be included in your statistical model (but is in scientific model!)
- Doing experiments is no protection against bad causal inference.

TABLE 1 Posttreatment Conditioning in Experimental Studies

Category	Prevalence
Engages in posttreatment conditioning	46.7%
Controls for/interacts with a posttreatment variable	21.3%
Drops cases based on posttreatment criteria	14.7%
Both types of posttreatment conditioning present	10.7%
No conditioning on posttreatment variables	52.0%
Insufficient information to code	1.3%

From Montgomery et al 2018 "How Conditioning on Posttreatment Variables Can Ruin Your Experiment and What to Do about It"

Unboserved Confound



The collider

Z is a collider



- X and Y are independent in the total sample; they share no cause $X \perp Y$.
- X and Y both influence Z.
- Once stratified by Z, X and Y are associated (not independent anymore): $X \not\perp Y|Z$. Stratifying by Z creates an association?!
- When we learn Z, we learn about the ranges of values of X and Y that could have produced this Z-value.

The collider: a binary example Without/with stratification

```
n <- 1000
X <- rbern(n, 0.5)
Y <- rbern(n, 0.5)
Z <- rbern(n,
      ifelse(X+Y>0, 0.9, 0.2))
table(X,Y)
```

```
##     Y
## X     0   1
##   0 259 261
##   1 245 235
```

```
cor(X,Y)
```

```
## [1] -0.01233026
```

```
table(X[Z==0], Y[Z==0])
```

```
##
##       0   1
##   0 207  15
##   1  24  26
```

```
cor(X[Z==0], Y[Z==0])
```

```
## [1] 0.4897983
```

```
table(X[Z==1], Y[Z==1])
```

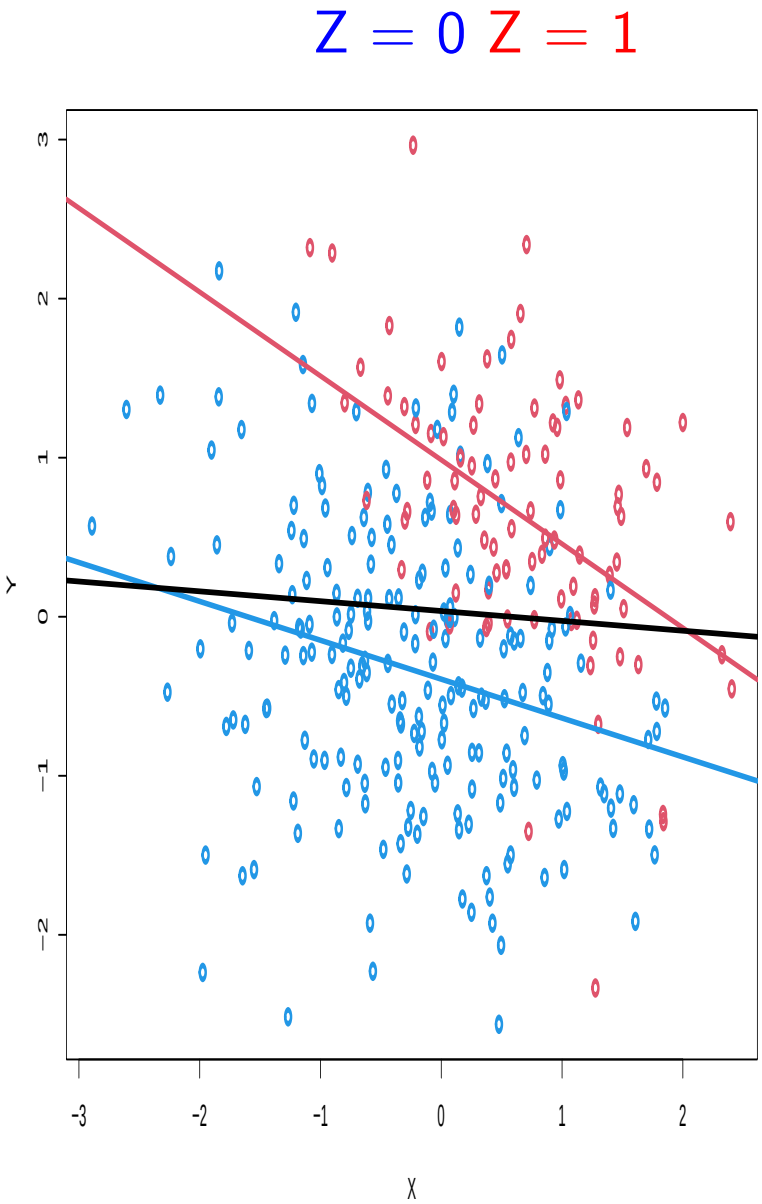
```
##
##       0   1
##   0  52 246
##   1 221 209
```

```
cor(X[Z==1], Y[Z==1])
```

```
## [1] -0.3447783
```

Continuous

```
cols <- c(4,2)
N <- 300
X <- rnorm(N)
Y <- rnorm(N)
Z <- rbern(N,
      inv_logit(2*X+2*Y-2))
```



Summary

- We have seen how to do multivariate regression modelling.
- Always start from the estimand, then state the scientific causal model and finally use appropriate regressions.
- Multiple predictors are useful for knocking out spurious associations.
- Another reason is to measure the direct influence of multiple predictors on an outcome. These influences can be not apparent from bivariate relationships; it occurs often, when two predictors are associated to each other, and one is "+"-associated with the outcome, whilst the other is "-"-associated with the outcome.
- A collider is not a confound.